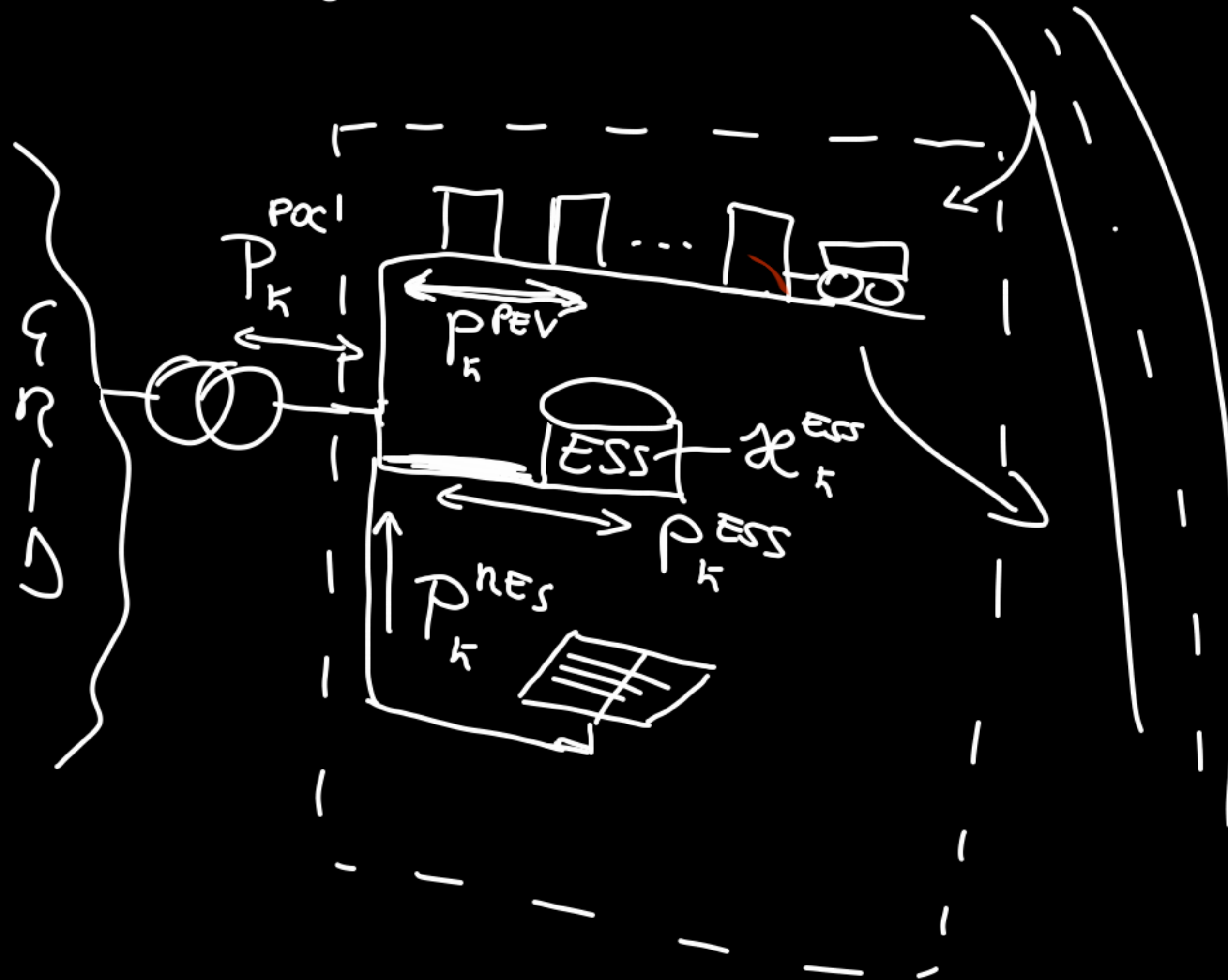
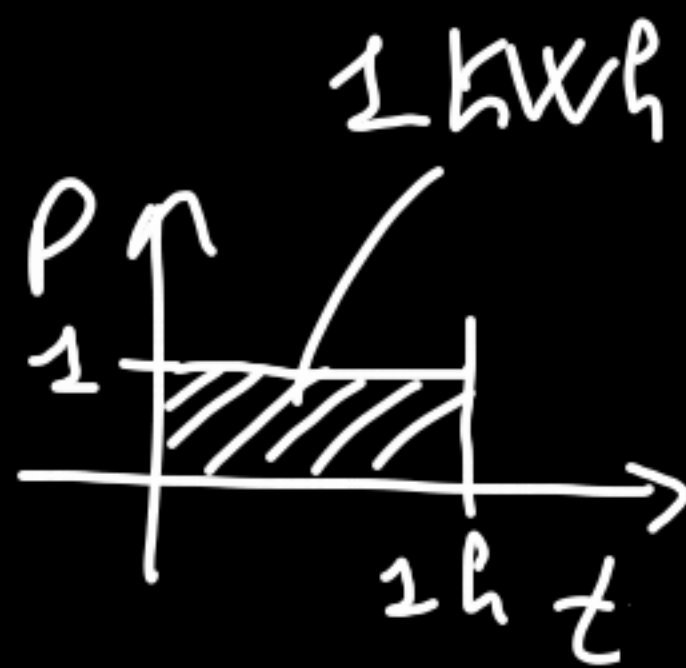


5/10/2026 - EXAMPLE ON CHARGING AREA CONTROL



- POWER [kW]
- ENERGY [kWh]



GOALS:

- PROVIDE MAXIMUM POWER TO PEVS (CONSTRAINT)



- KEEP P_k^{POC} CLOSE TO 0 kW (REGULATION)



MPC FORMULATION

TARGET
FUNCTION

$$\min \left\{ \underbrace{\sum_{i=0}^{N-1} \left[q \left(p_{ilk}^{POC} \right)^2 + b \left(x_{ilk} - x^{ref} \right)^2 \right]}_{\ell(\cdot)} + \underbrace{b^+ \left(x_{N|k} - x^{ref} \right)^2}_{V_f(x_{N|k})} \right\}$$

CONSTRAINTS

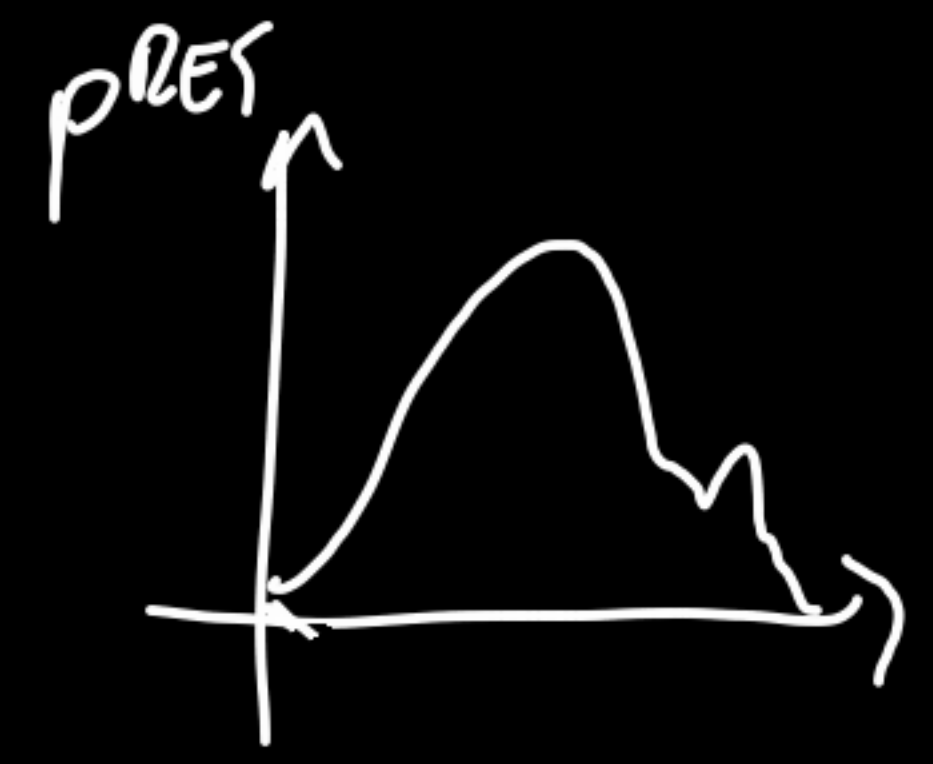
$$y_k = C x_k$$

$\swarrow \searrow$
 $y_k^T S y_k \quad x_k^T \overbrace{C^T S C} \quad x_k$

OUTPUT
REGULATION

- BALANCE CONSTRAINT (POWER BALANCE CONSTRAINT / GLOBAL CONSTRAINT)
COUPLING CONSTRAINT

$$p_{ilk}^{POC} = p_{ilk}^{ESS} + p_{ilk}^{PEV} - p_{ilk}^{RES}, \quad i=0, 1, \dots, N-1$$



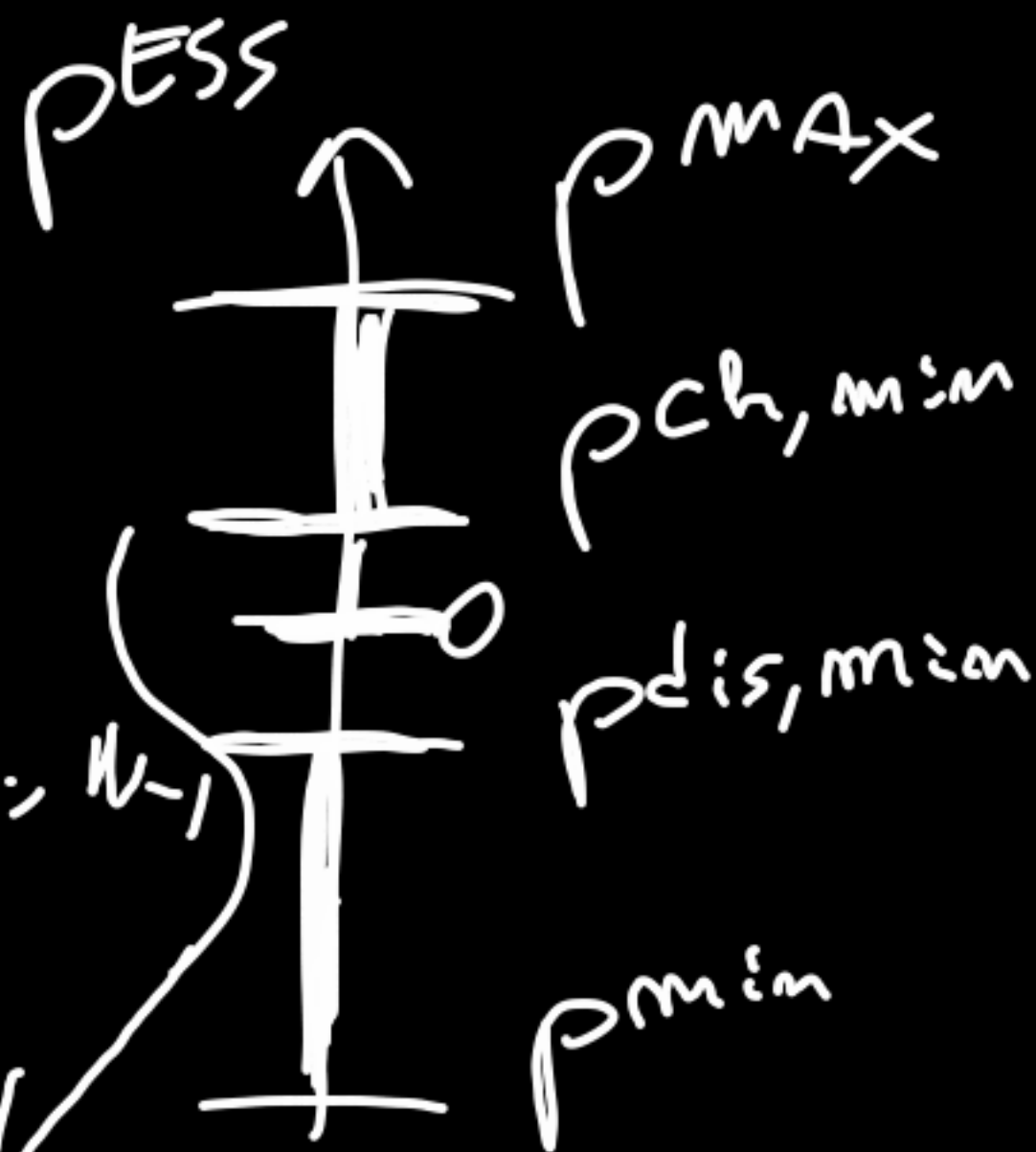
TRIVIAL CONSTRAINTS...

BOX CONSTRAINTS

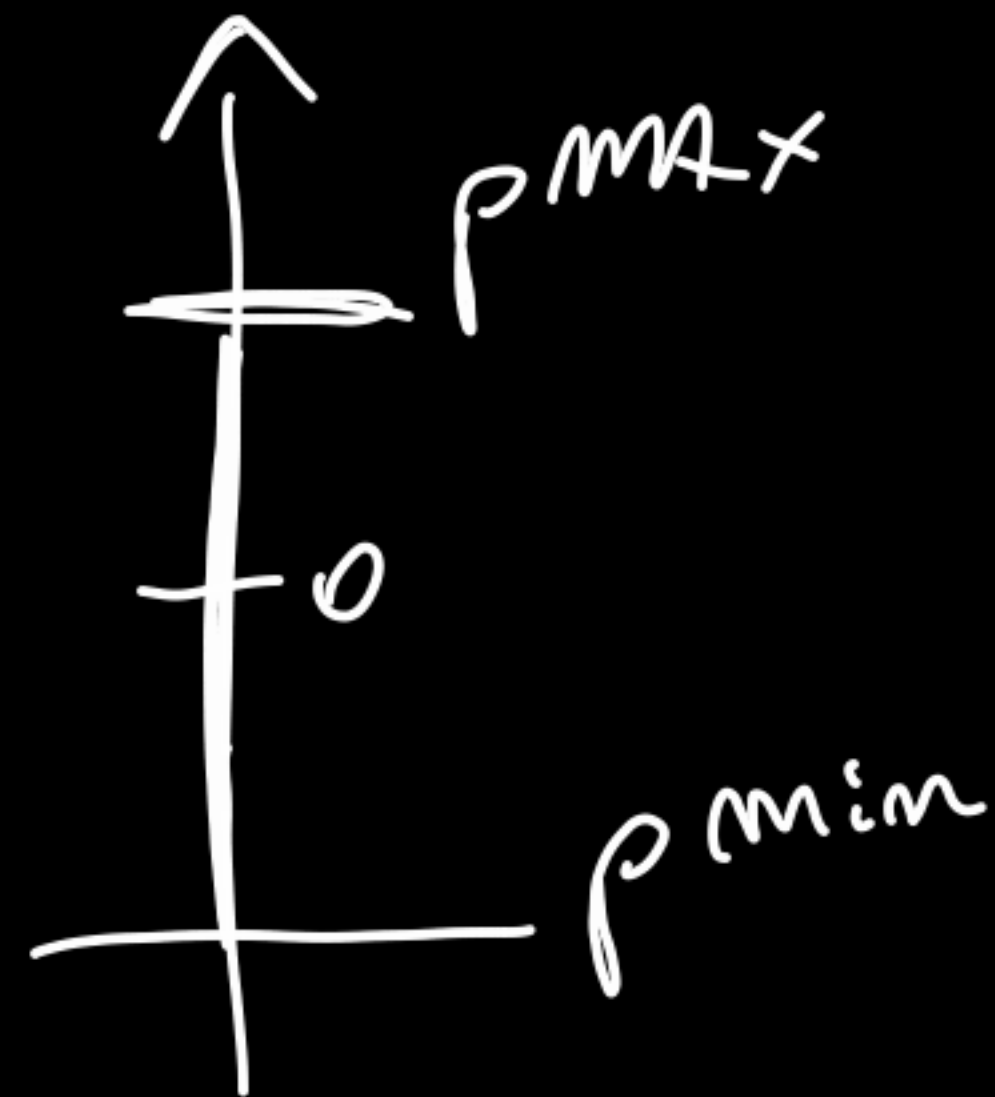
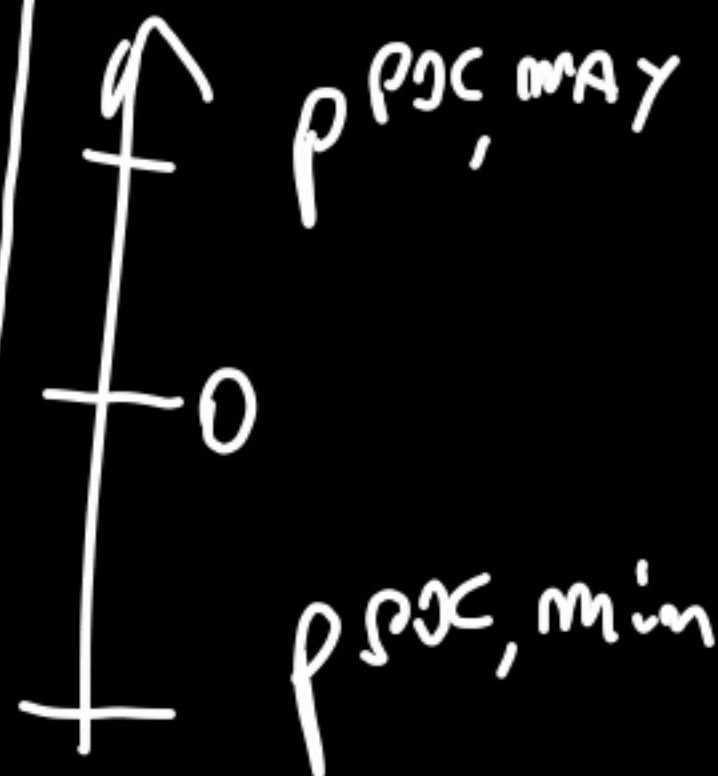
$$p^{\min} \leq p_{ik}^{\text{ESS}} \leq p^{\max}, \quad i=0, \dots, N-1$$

$$x^{\min} \leq x_{ik}^{\text{ESS}} \leq x^{\max}, \quad i=0, \dots, N$$

$$p^{\text{POC}, \min} \leq p_{ik}^{\text{POC}} \leq p^{\text{POC}, \max}, \quad i=0, 1, \dots, N-1$$



SEMI-CONTINUOUS
VARIABLE



CONT. VARIABLE

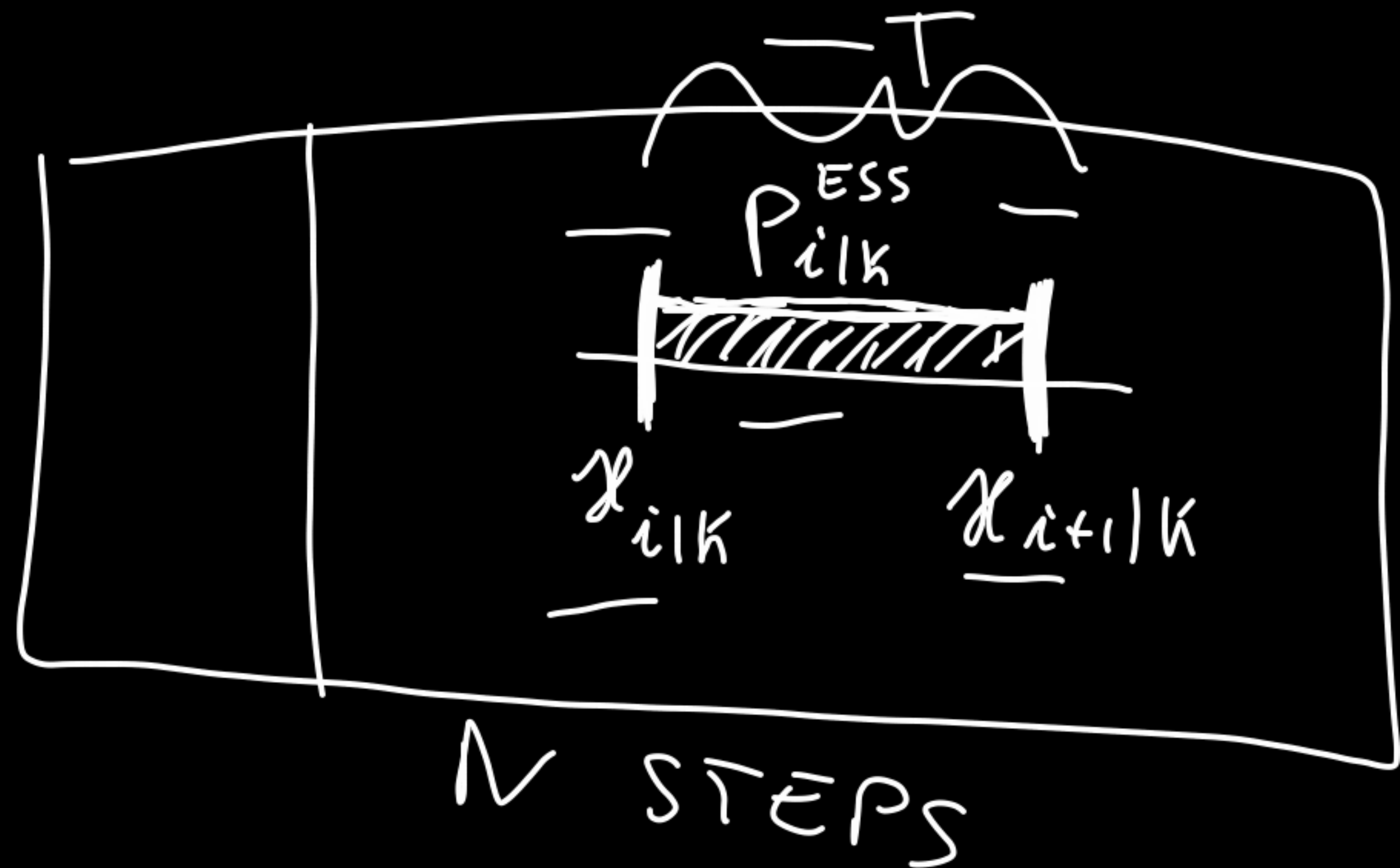
CONSTRAINTS

INITIAL STATE

$$x_{0|k} = x_k \text{ (KNOWN)}$$

$$x_{i+1|k} = x_{i|k} + \underbrace{T}_{\substack{\uparrow \\ T=1\text{ min} \rightarrow \frac{1}{60}\text{ h}}} \cdot P_{i|k}^{\text{ESS}}, \quad i=0, \dots, N-1$$

$$T = 1 \text{ min} \rightarrow \frac{1}{60} \text{ h}$$

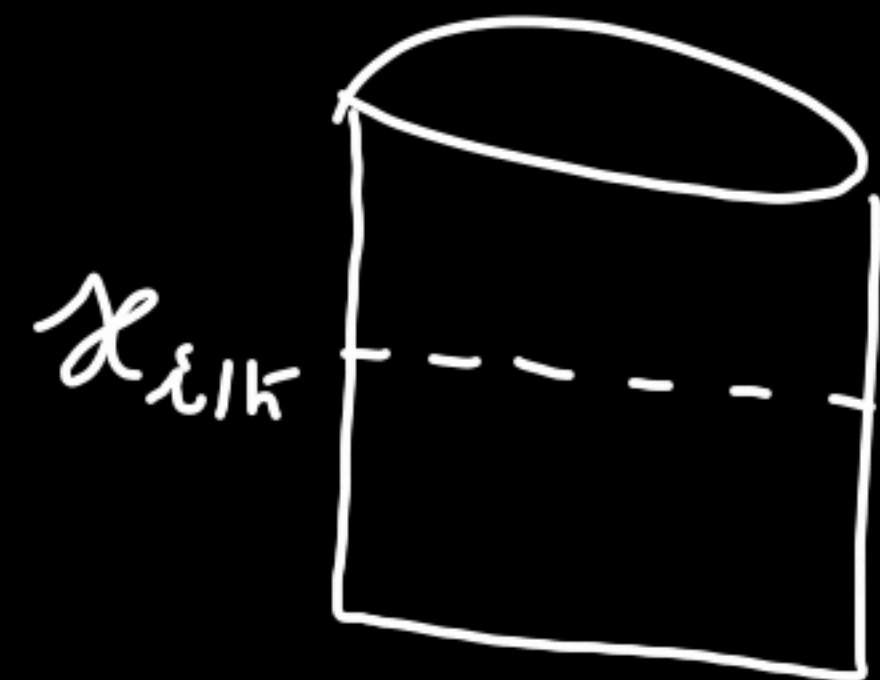


NO LOSSES

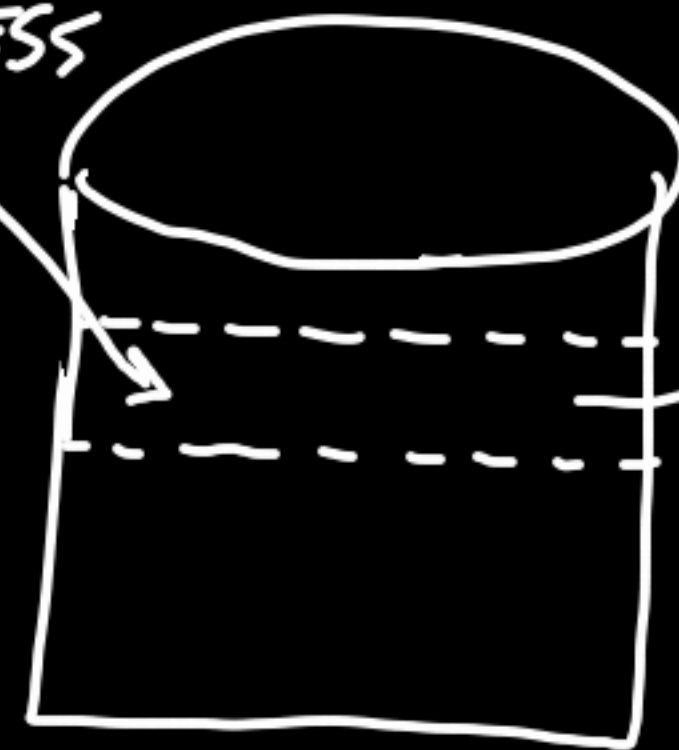
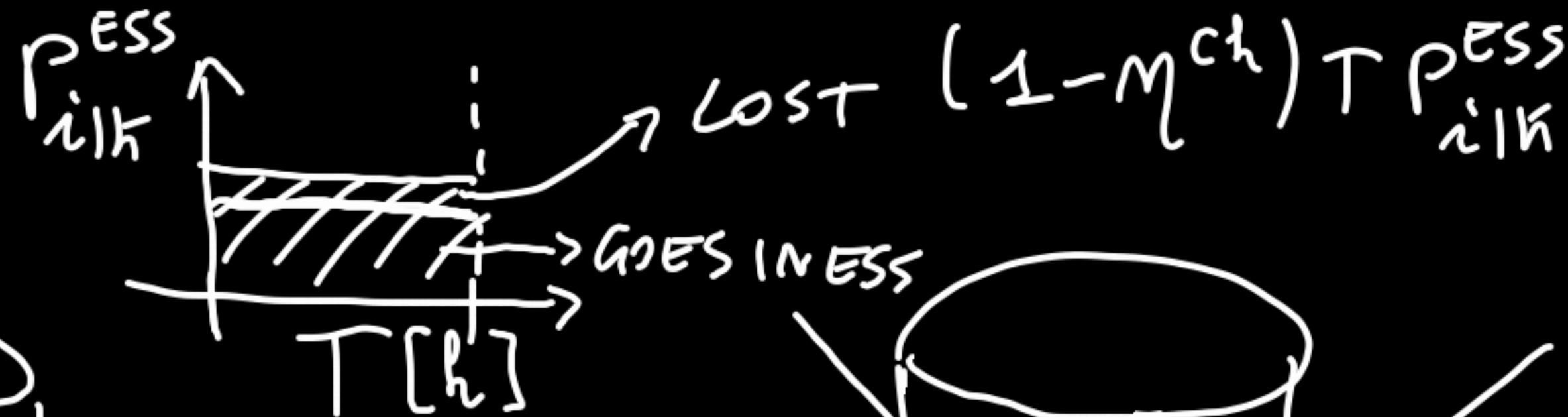
ESS MODEL WITH LOSSES → MILP PROBLEM

CHARGING LOSSES

$$P_{ik}^{ESS} > 0$$



i/k



$i+1/k$

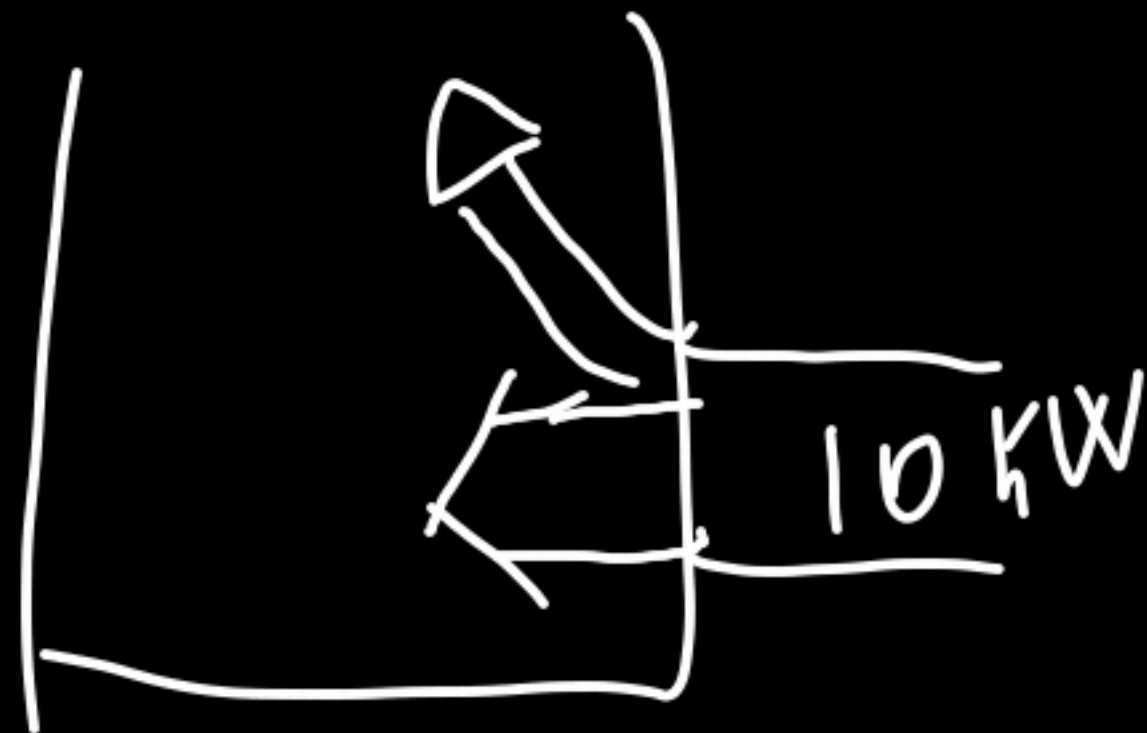
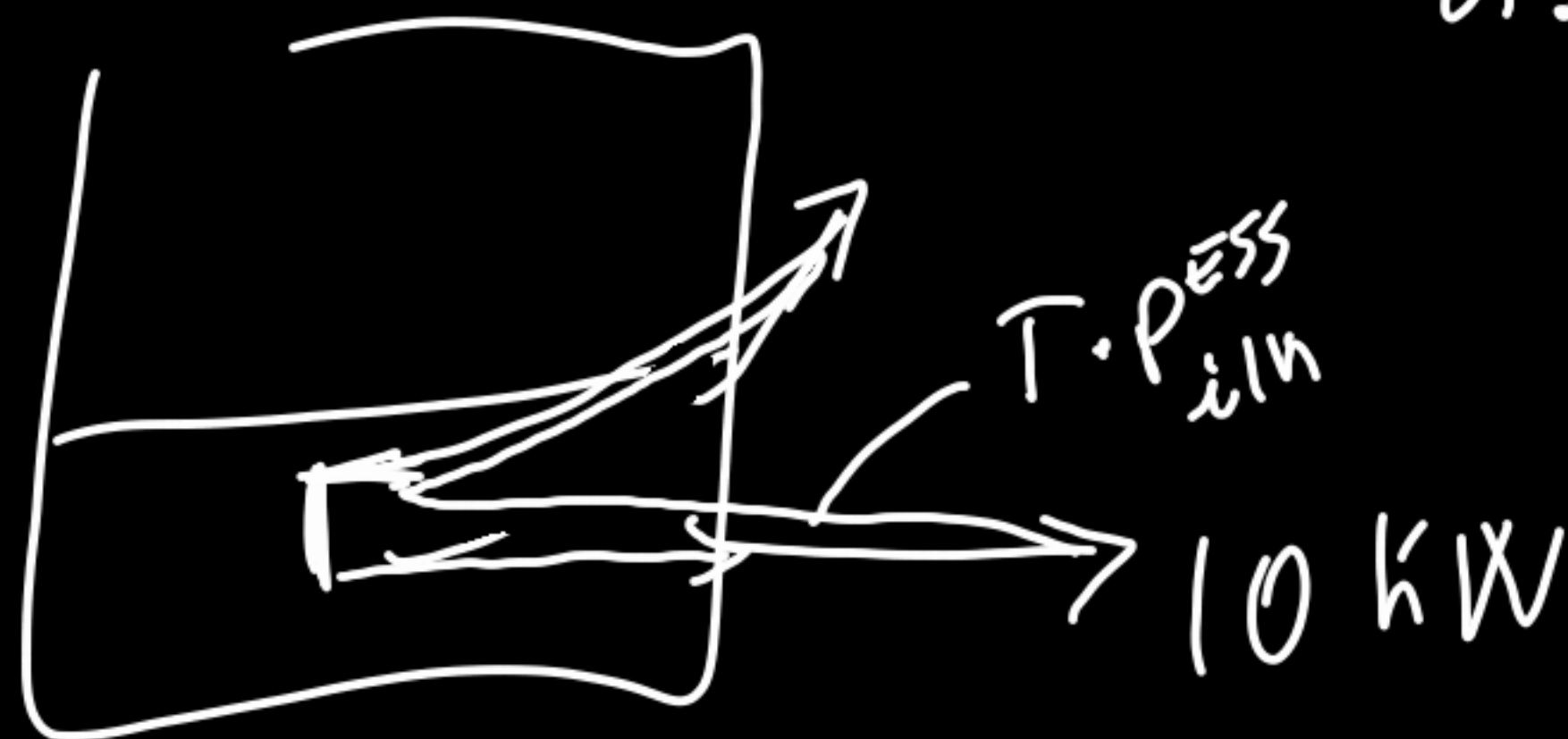
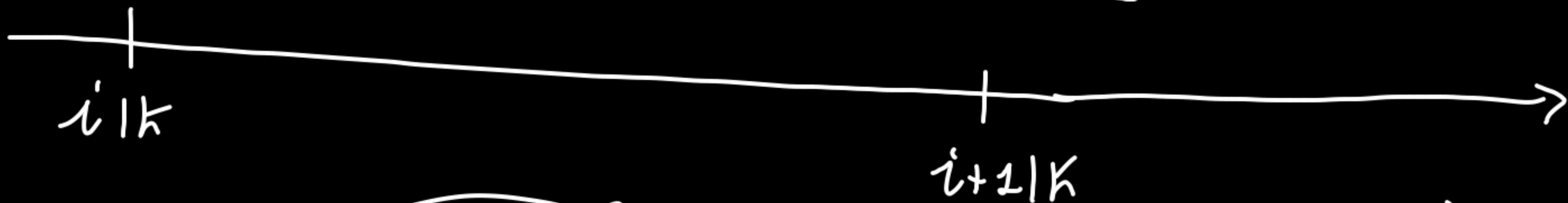
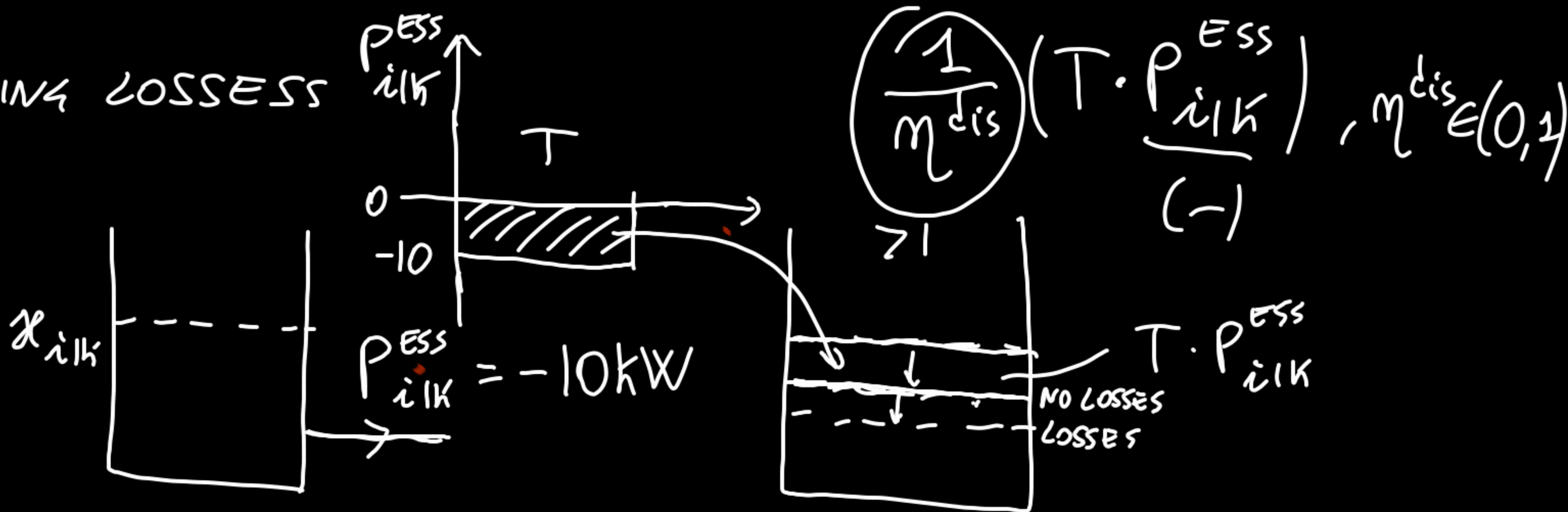
$$\eta^{ch} (T \cdot P_{ik}^{ESS})$$

$$\eta^{ch} \in (0, 1)$$

η^{ch} TYPICALLY NOT CONSTANT

DISCHARGING LOSSES

$$P_{i|k}^{ESS} < 0$$



ESS MODEL WITH LOSSES

$$x_{i+1,k} = x_{i,k} + \underbrace{T \cdot \eta^{ch} \cdot \max\{0, P_{i,k}^{ESS}\}}_{\text{CHARGING}} + \underbrace{T \cdot \frac{1}{\eta^{dis}} \cdot \min\{0, P_{i,k}^{ESS}\}}_{\text{DISCHARGING}} \quad i=0, \dots, N-1$$

- 0 IF $P_{i,k}^{ESS} < 0$

- $T \eta^{ch} P_{i,k}^{ESS}$ IF $P_{i,k}^{ESS} > 0$

- 0 IF $P_{i,k}^{ESS} > 0$

- $T \cdot \frac{1}{\eta^{dis}} P_{i,k}^{ESS}$ IF $P_{i,k}^{ESS} < 0$

NONLINEAR EQUATIONS

EQUIVALENT LINEAR MODEL

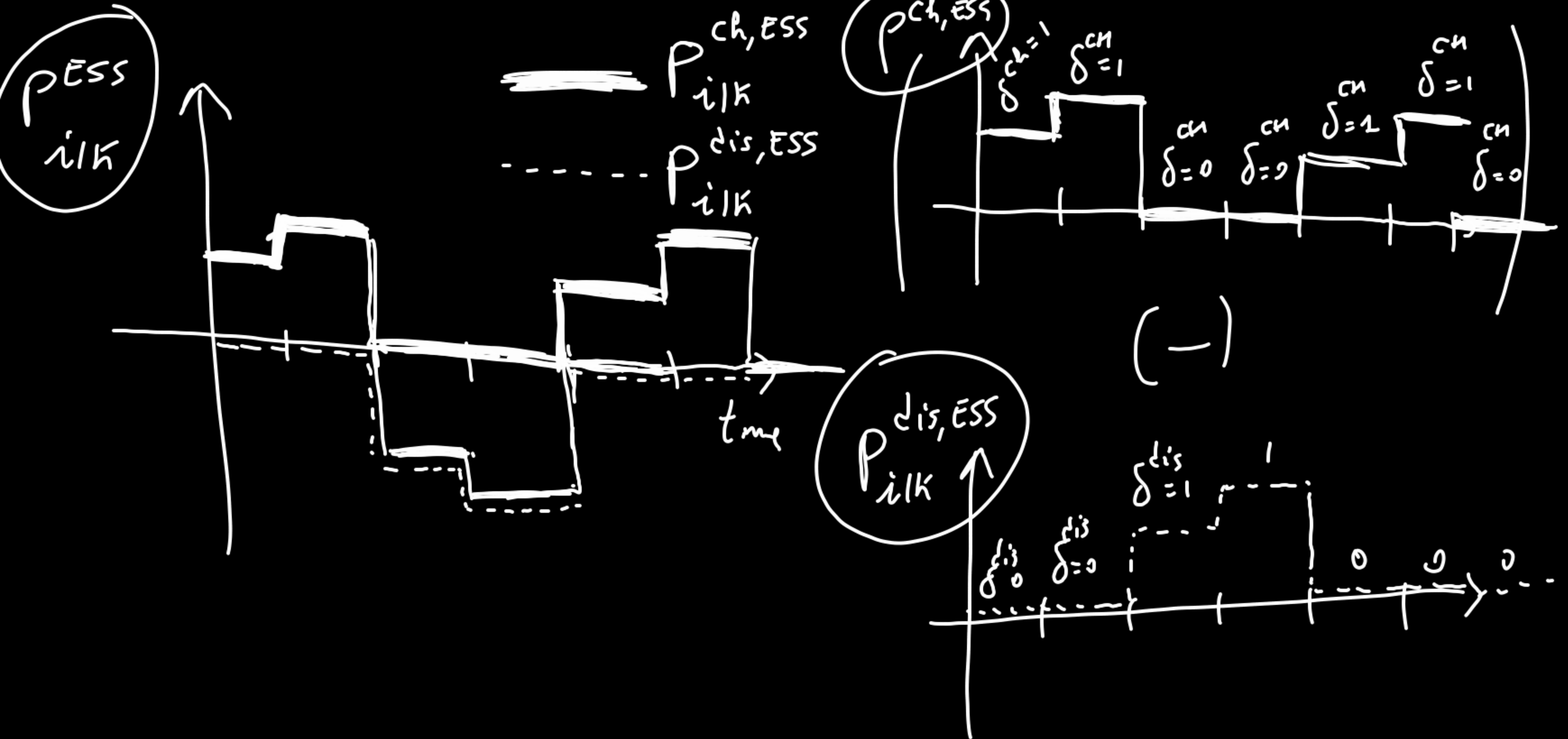
PRICE TO PAY : ADD BOOLEAN VARIABLES

→ CONTINUOUS PROGRAMMING → MIXED-INTEGER
LINEAR PROGRAMMING
(MILP)

BUT:

MILP SIMPLER THAN NONLINEAR PROGRAMMING

DECOMPOSE P_{ilk}^{ESS} AS THE ALGEBRAIC SUM OF CHARGING POWER AND A DISCHARGING POWER.



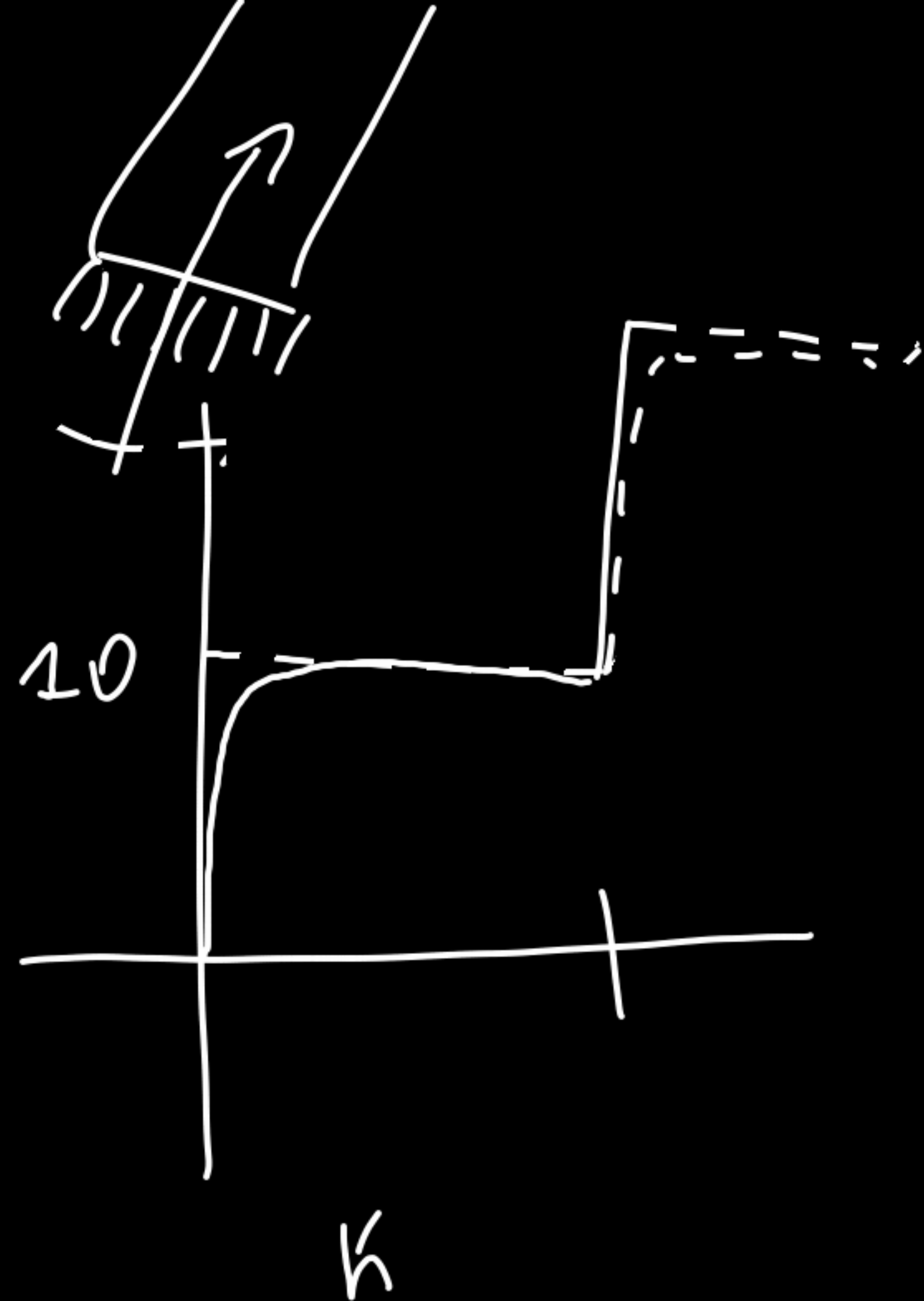
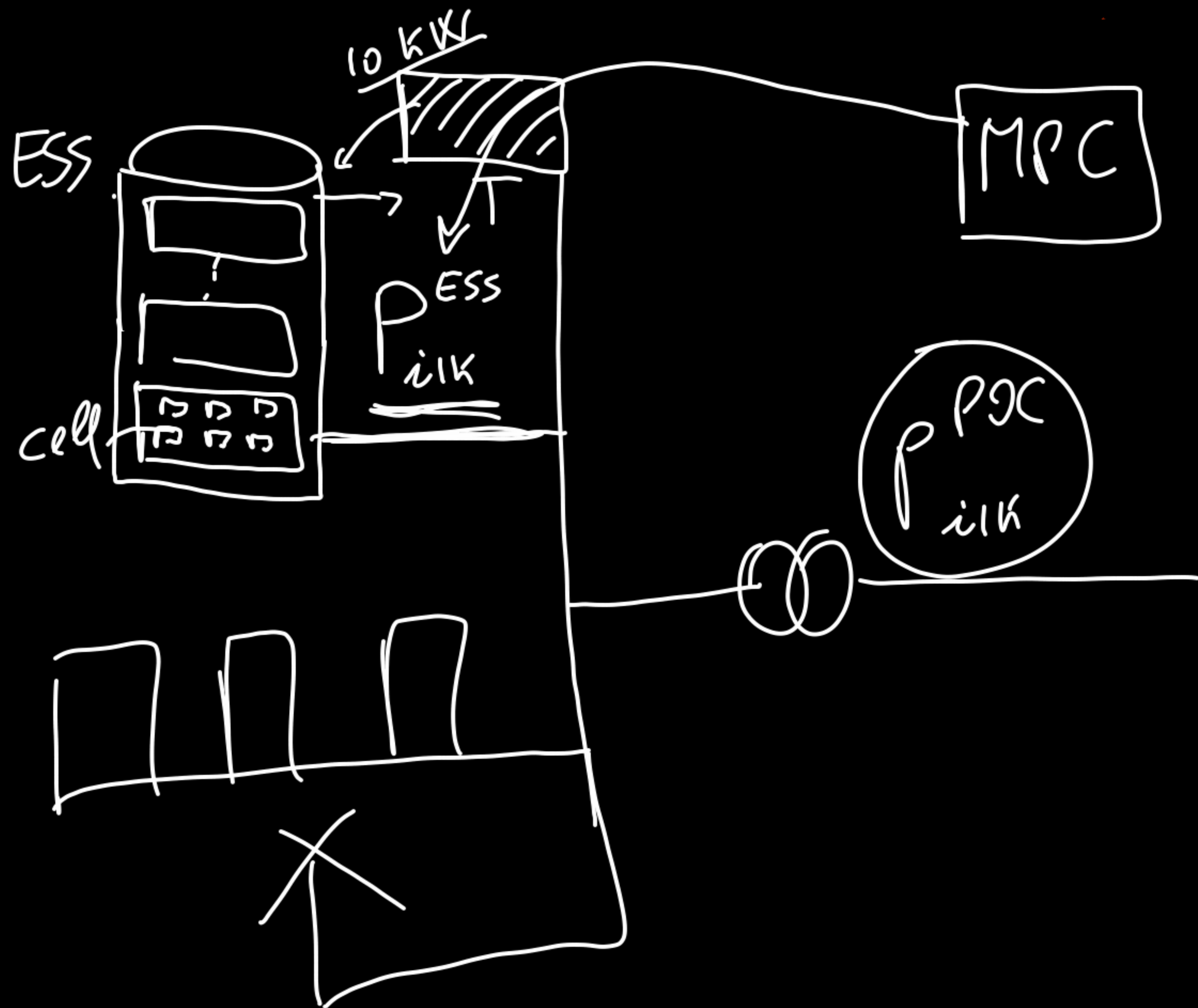
ADD VARIABLES

- $P_{i|k}^{ch,ESS} \geq 0, i = 0, 1, \dots, N-1$

- $P_{i|k}^{dis,ESS} \geq 0, i = 0, 1, \dots, N-1$

- $P_{i|k}^{ESS} = P_{i|k}^{ch,ESS} - P_{i|k}^{dis,ESS}, i = 0, 1, \dots, N-1$

SMALL NOTE ON HIERARCHICAL CONTROL



SPLIT OF p_{ik}^{ESS}

$$p_{ik}^{ch, ESS}, p_{ik}^{dis, ESS} \geq 0, i = 0, \dots, N-1$$

$$p_{ik}^{ESS} = p_{ik}^{ch, ESS} - p_{ik}^{dis, ESS}, i = 0, 1, \dots, N-1$$

$$0 \leq p_{ik}^{ch, ESS} \leq p_{ESS, max}, i = 0, 1, \dots, N-1$$

$$0 \leq p_{ik}^{dis, ESS} \leq p_{ESS, dis}$$

$p_{ik}^{ch, ESS}$, $p_{ik}^{dis, ESS}$

LOGICAL CONSTRAINT: AT ANY GIVEN TIME, THEY CANNOT BE BOTH POSITIVE.

MUTUALLY EXCLUSIVE.

EVENT 1. ESS CHARGING AT $i|k$ ($p_{ik}^{ESS, ch} > 0$).

EVENT 2. ESS DISCHARGING AT $i|k$ ($p_{ik}^{dis, ESS} > 0$).

WE CAN INTRODUCE ONE "INDICATOR VARIABLE"
(BOOLEAN VARIABLE) FOR EVERY EVENT.

$$\delta_{i|k}^{ch} \in \{0, 1\}, \forall_{i=0, \dots, N-1} \left[\delta_{i|k}^{ch} = 1 \iff (P_{i|k}^{ch, ESS} > 0) \right]$$

$\delta_{i|k}^{ch} = 1$ INDICATES THAT AT $i|k$ ESS IS CHARGING

BEHAVIOUR I WANT TO IMPOSE TO VARIABLE

$$\delta_{i|k}^{ch}$$

- $\delta_{ik}^{\text{dis}} \in \{0, 1\}, i = 0, \dots, N-1$ WHEN $= 1$, INDICATES THE ESS IS DISCHARGING (EVENT 2 IS TRUE).

$$\left(\delta_{ik}^{\text{dis}} = 1 \iff p_{ik}^{\text{dis, ESS}} > 0 \right)$$

|||

WE DO THIS (? MATHEMATICAL CONSTRAINT BETWEEN
LATER δ_{AN}^{dis} p^{dis})

CONSTR. BETWEEN δ_{ik}^{ch} AND δ_{ik}^{dis} TO ENSURE
CHARGING (EVENT 1) AND DISCHARGING (EVENT 2)
ARE MUTUALLY EXCLUSIVE.

$$\delta_{ik}^{ch} + \delta_{ik}^{dis} \leq 1$$

$$, i = 0, 1, \dots, N-1$$

EVENT 1

$$\delta_{ik}^{ch} = 1$$

EVENT 2

$$\delta_{ik}^{dis} = 1$$

